Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the test accuracy during training.

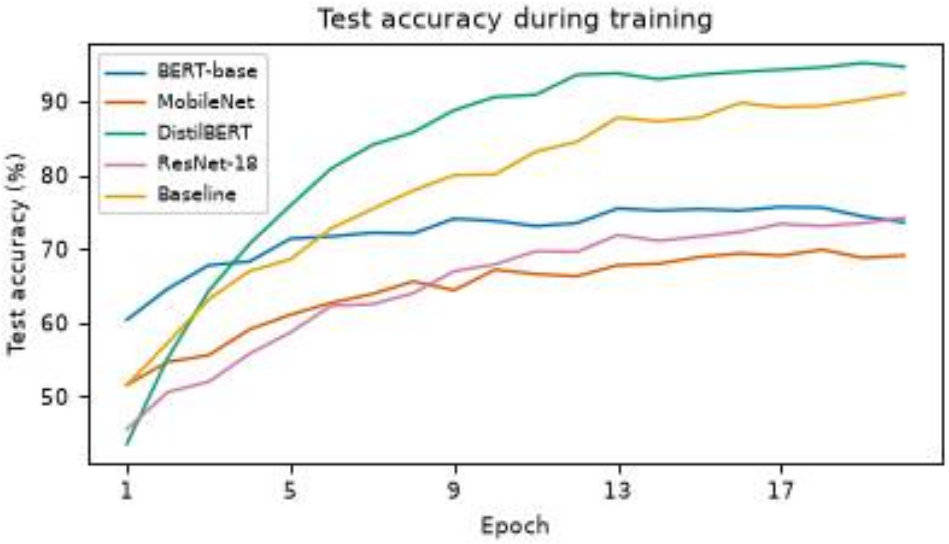


Figure 1: Test accuracy during training for the models in this experiment.

From epoch 3 to epoch 20, the test accuracy of ResNet-18 improves by a relative about 40%. From epoch 5 to epoch 15, the test accuracy of ResNet-18 improves by a relative 70.5%.

Figure 2 shows the validation accuracy during training.

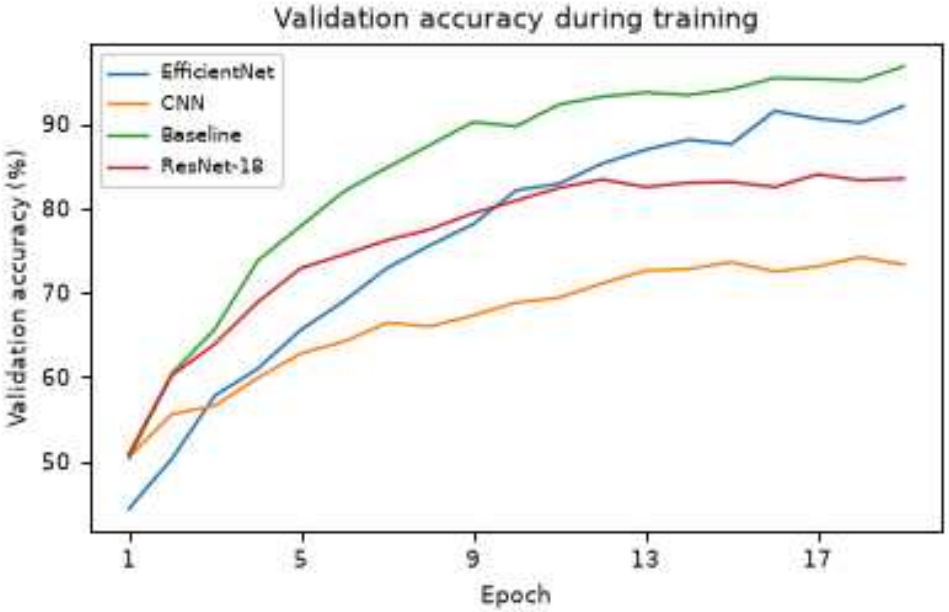


Figure 2: Validation accuracy during training for the models in this experiment.

Baseline outperforms EfficientNet by 15.7 percentage points at epoch 4. As shown in Figure 2, CNN attains a validation accuracy of 61.1% at epoch 13.

3 Discussion

We note that the training was stable for all models.